

# Zixun Huang

 [zixunh@wharton.upenn.edu](mailto:zixunh@wharton.upenn.edu) —  [Homepage](#) —  [GitHub](#) —  [Google Scholar](#) —  [LinkedIn](#)

## RESEARCH INTERESTS

Theoretical foundations of machine learning, with an emphasis on **optimization theory and scaling laws**, **LLM reasoning and post-training**, and **applications of AI**.

## EDUCATION

|                                                                                                                                                            |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>The Wharton School, University of Pennsylvania</b><br><i>Ph.D. in Statistics and Data Science</i>                                                       | Aug. 2026 – Present    |
| <b>Peking University, School of Mathematical Sciences</b><br><i>B.S. in Statistics</i> ; Elite Undergraduate Program for Applied Mathematics; GPA: 3.7/4.0 | Sept. 2022 – Jun. 2026 |
| <b>University of California, Berkeley</b><br><i>Visiting Student</i> ; GPA: 4.0/4.0                                                                        | Jan. 2025 – Aug. 2025  |

## WORKING EXPERIENCE

|                                                                            |                       |
|----------------------------------------------------------------------------|-----------------------|
| <b>Tencent Hy – Qingyun Talent Program</b> <i>Research Intern, Beijing</i> | Jun. 2026 – Aug. 2026 |
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## PUBLICATIONS

- [1] **Zixun Huang\***, Jiayi Sheng\*, Zeyu Zheng. *Variance-Aware Baselines and Adaptive Learning Rates for Reinforcement Learning with Verifiable Rewards*. **Under review**. [\[Paper\]](#)
- [2] Yuhao Sun, Zekun Wu, **Zixun Huang**, Peijie Zhou. *TracingFlow: A Simulation-Free Trajectory Inference Framework Based on Second-Order Dynamics*. **Under review**.
- [3] Binghui Li\*, Fengling Chen\*, **Zixun Huang\***, Lean Wang\*, Lei Wu. *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*. **NeurIPS 2025 Spotlight**. [\[Paper\]](#) \* Equal contribution.

## RESEARCH EXPERIENCE

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| <b>University of California, Berkeley</b> <i>Research Intern, Berkeley, CA</i><br><i>Supervisor: Associate Professor Zeyu Zheng</i> | Jan. 2025 – Aug. 2025 |
| <b>Peking University</b> <i>Research Intern, Beijing</i><br><i>Supervisor: Assistant Professor Lei Wu</i>                           | May 2023 – Aug. 2026  |

## SELECTED HONORS AND AWARDS

|                                             |      |
|---------------------------------------------|------|
| Grand Prize, “Challenge Cup” Competition    | 2026 |
| Weiming Scholar, Peking University          | 2026 |
| Hong Sheng Scholarship                      | 2024 |
| First Prize, Chinese Mathematics Contest    | 2024 |
| Yau Contest Groups Prize                    | 2024 |
| Xiaomi Scholarship                          | 2023 |
| Gold Prize, Chinese Mathematical Olympiad   | 2021 |
| Silver Prize, Chinese Mathematical Olympiad | 2020 |

## SKILLS

|                     |                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------|
| <b>Programming:</b> | Python (NumPy, pandas, scikit-learn, PyTorch), C++, MATLAB, L <sup>A</sup> T <sub>E</sub> X |
| <b>Language:</b>    | GRE V160/Q170; GRE Math 970 (95th percentile); TOEFL 101                                    |